

PROBABILITY 3: PROBLEMS DISCUSSED ON SEPT 1

Solve the following problems using MCT, DCT or Fatou's lemma. In what follows, we fix $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space, and we will denote events by $A \in \mathcal{F}$ and random variables by X .

1. Let $A_1 \subset A_2 \subset \dots$ be an increasing family of events in $\mathcal{F}$ such that $\cup_{i \geq 1} A_i = \Omega$. Let $X \geq 0$ be a r.v., show that

$$\mathbb{E}[X \cdot \mathbb{1}_{A_n}] \rightarrow \mathbb{E}[X].$$

2. Let $X \geq 0$ be a r.v. taking integer values, show that

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

3. Suppose $0 \leq X \leq 1$, then what is $\lim_{n \rightarrow \infty} \mathbb{E}[X^n]$?

4. Markov's inequality: Let $X \geq 0$, then show that for every $t > 0$:

$$t\mathbb{P}(X \geq t) \leq \mathbb{E}[X].$$

5. Suppose $X \geq 0$ and $\mathbb{E}[X] < \infty$, show that

$$\lim_{t \rightarrow \infty} t\mathbb{P}(X \geq t) = 0.$$

6. Suppose $X \geq 0$ is such that $\lim_{t \rightarrow \infty} t\mathbb{P}(X \geq t) = 0$. Does this imply $\mathbb{E}[X] < \infty$?

7. Borel-Cantelli 2: Suppose A_n 's are such that $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$. Then $\mathbb{P}(A_n \text{ i.o.}) = 0$.